WIRTSCHAFTS UNIVERSITÄT WIEN VIENNA UNIVERSITY OF ECONOMICS AND BUSINESS

Spatial Housing Inequality and Supply Inelasticity in Germany

Master's Thesis

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A.1 Classification of German cities as Superstar housing markets across five-year periods from 2000 to 2024. Panel A applies the static identification strategy, where inelasticity thresholds are compared with the sample median. Panel B applies period-specific identification, where thresholds are computed using two-period windows. Cities marked in blue are classified as Superstars. The training window 1995-1999 is excluded from the visualization. 28

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Abstract

This thesis examines whether housing supply constraints explain the growing divergence in housing prices across German cities. Building on the "Superstar cities" framework of Gyourko et al. (2013), it tests whether cities characterized by strong demand and inelastic housing supply exhibit persistently higher housing prices and different income dynamics. Using city-level data on housing prices, construction activity, and income, the analysis applies panel regression models to apartment and single-family housing markets. The results show that Superstar cities consistently experience higher housing prices, while evidence for income sorting is weak once city fixed effects are included. Overall, constrained supply appears central to explaining spatial housing inequality in Germany.

1. Introduction

Since the last quarter of the 20th century there has been a divergence in the cost of housing with an increasing difference across cities. While some locations have experienced modest and stable increases, others have undergone sustained and significantly higher price growth over long periods. Although a majority of studies focus on the United States, e. g. in Glaeser et al. (2012) and Gyourko et al. (2013), similar spatial disparities have been documented across several developed economies. Hilber and Vermeulen (2016) provide evidence for the United Kingdom, Büchler et al. (2021) analyze the Swiss case. Despite this growing body of evidence, comparatively little is known about whether the mechanisms underlying these divergences, particularly the interaction between housing supply constraints and local demand, translate into similar outcomes in other institutional settings such as Germany.

In Germany, the evolution of housing markets since reunification illustrates this pattern particularly well. The early 1990s were characterized by a sharp increase in housing demand, driven by large migration flows from East to West following the fall of the Berlin Wall. This resulted in a significant housing shortage in many parts of western Germany, which prompted new government support in terms of tax benefits and direct subsidies allowing households to buy new houses and incentivizing developers to expand supply. According to the German Bundesbank, from 1991 to 1995 the number of completed residential units increased by more than 16% year-on-year.[1] However, this period of expansion seems an exception if one considers the following 20 years where new construction decreased on average by 8% every year.

Although price growth has been particularly modest throughout the first decade of the 21st century, it rose sharply from 2010 and remained steadily at an average of 7% growth year-on-year from 2014 onward.[2] Figure 1.1 shows year-on-year percentage changes in new residential completions and house prices at the national level. Construction shows a clear cyclical component with large swings that change through the years. The early 1990s building push is quite evident, and one can observe the sharp decline which lasted until 2010. Price-wise, there has been a steady increase of the growth rate, that has remained at high levels despite the years before and during the Covid pandemic even though a decline followed shortly after. While it is true that lower house price growth was supported by the 1990s investment push, the Bundesbank observes in its June 2010 Monthly Report that lower demand for housing is also a result of bad fundamentals. Demographic stagnation and low economic growth throughout the 2000s played a key role in reducing house values.

Crucially, these developments were not uniform across space. When it comes to location decisions, households rarely look at the national macroeconomic outlook. The choice falls choose between specific regions and cities, where economic conditions and housing markets

[1] https://www.bundesbank.de/dynamic/action/en/statistics/time-series-databases/time-series-databases/745582/745582?listId=www_siws_iswi_realw1&tsTab=2&tsId=BBDY1.A.B30.N.G700.R0010.A&id=0 retrieved on 10/02/2026

[2] https://www.bundesbank.de/dynamic/action/en/statistics/time-series-databases/time-series-databases/745582/745582?tsTab=1&listId=www_siws_iswi_preise1&tsId=BBDY1.A.B10.N.G100.P0020.A&id=0 retrieved on 10/02/2026

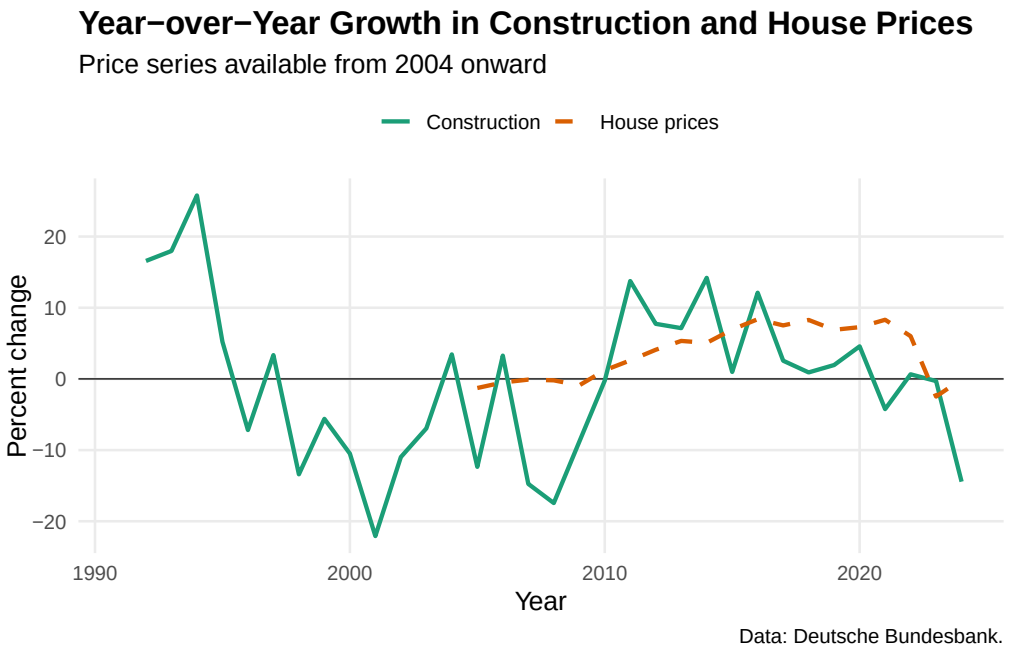

Figure 1.1.: The figure reports year-over-year percentage changes in residential construction values and the house price index in Germany. The house price data are available from 2004 onward, which explains the shorter series. Data: Deutsche Bundesbank.

can differ substantially. This raises the question of how housing price dynamics have evolved across German cities over the past three decades.

The largest spatial gap has historically been between eastern and western regions. Economic stagnation and worse employment conditions forced many households to relocate from Eastern to Western Germany. Furthermore, not all western cities experienced the same immigration pressure. Already by 1995, many regions that underwent the post-WWII economic miracle (Wirtschaftswunder) were already enduring a period of economic slowdown. In terms of GDP growth, cities such as Duisburg, Bochum, and Dortmund performed similarly to their eastern counterparts like Chemnitz or Weimar, and this was also reflected in real estate price dynamics. Figure 1.2 reports price level indexes and distributions for residential dwellings from a sample of German cities. From panel A, one can observe price dynamics from 1995. After the initial rise from stronger demand due to population movements, there has been a general decline reflecting the weak fundamentals mentioned above. After 2010, this pattern changed markedly. Demand started to pick up again and some cities began to experience persistently higher prices. Interestingly, all cities eventually saw an increase in house prices. A common explanation points to inelastic housing supply that has not kept pace with rising demand. This idea may help explain why there has been such a large gap in the timing of rising prices. Individuals opt to move to more peripheral cities (in terms of local productivity) once housing costs become unaffordable in cities where economic activity is more concentrated to offset lower wages with lower expenditure. To better show this sharp increase in dispersion, Figure 1.3 reports the distributions in house prices from 1995 and 2020. The plot not only suggests that average prices increased, but also that variation increased too; most of the density

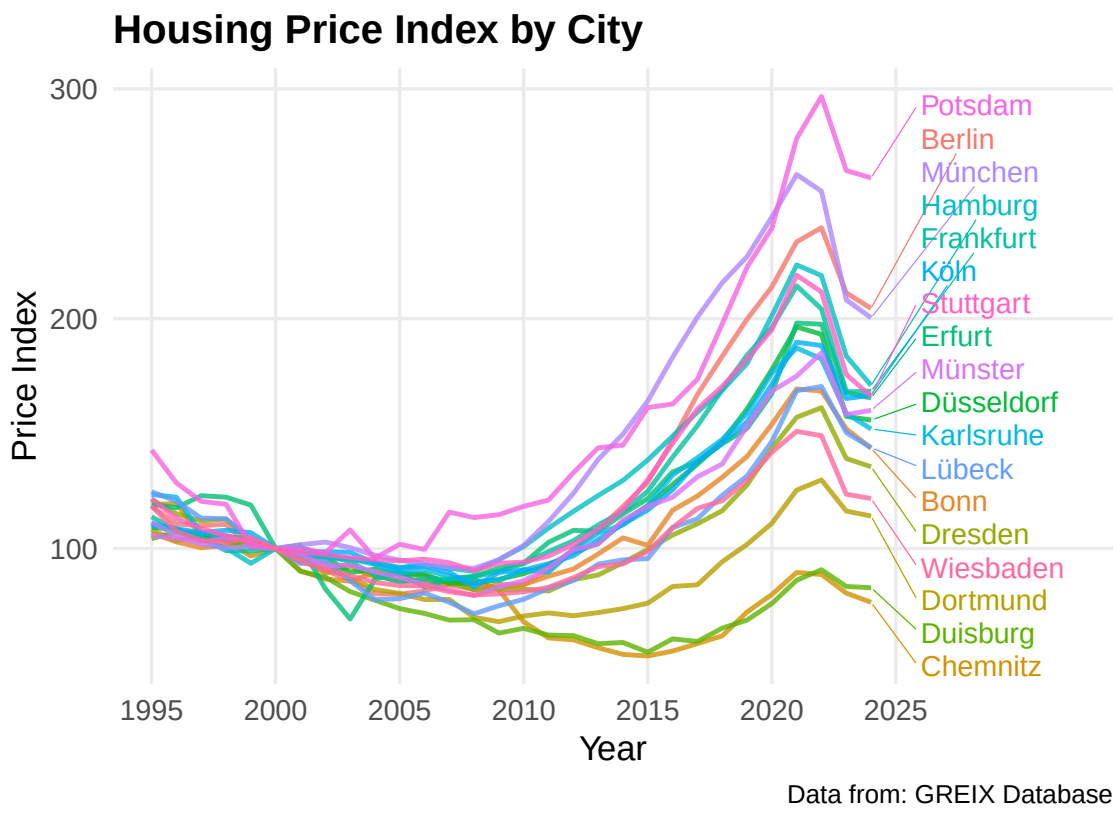

Figure 1.2.: The evolution of the housing price index between 1995 and 2024. For readability, I select a random sample of half of the cities. The index is normalized at value 100 for the year 2000 for each city. Lines are labeled at the last available year.

in 2020 is concentrated to the right of the 1995 distribution, while some cities are still recovering in 2020 their 1995 levels.

This raises the question of what drives this sharp increase in spatial dispersion. Is there a large gap between the demand for housing and its supply? As the average household becomes more skilled and therefore more productive, even if location preferences are held constant over time, one would expect higher demand for housing in the largest cities, which offer more opportunities and better amenities. If this is the case, then supply should adjust accordingly, and such a large spatial divergence in price growth appears puzzling.

One way to formalize these dynamics is provided by the "Superstar cities" framework developed by Gyourko et al. (2013). Their work proposes a mechanism that accounts for dispersion in house price growth rate in the long run without relying on persistent changes in local fundamentals or people's tastes. They argue that the cause of most price dispersion is based on demand and supply for housing. Demand is driven by factors such as population growth, rising incomes, and the concentration of economic activity, while supply is shaped by geographic constraints, regulatory restrictions, existing urban density, or high construction costs. Cities that combine strong demand with inelastic supply are classified as "Superstars." In these locations, increases in demand are capitalized primarily into higher housing prices rather than into proportional increases in the housing stock. As a result, Superstar cities are characterized by persistently high and growing housing costs,

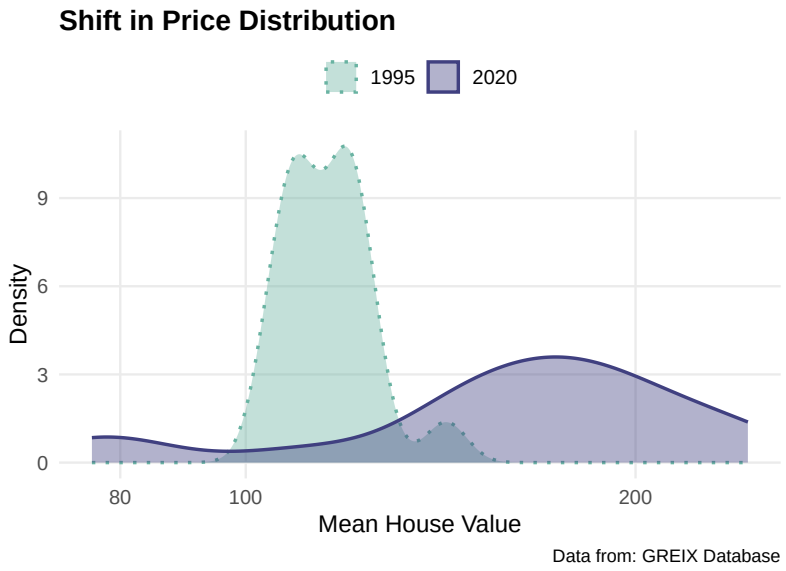

Figure 1.3.: The plot reports kernel density estimates of mean house values in 1995 and 2020, plotted on a logarithmic scale. Data are from the GREIX database.

as well as sorting of households who are willing and able to pay a premium to move to these locations.

This thesis applies the Superstar cities framework to the German context and tests its main empirical predictions. I find that cities classified as Superstars exhibit housing prices that are approximately 18% higher than in other cities. This difference remains statistically significant, though smaller at around 10%, once controlling for city fixed effects. In contrast, the evidence on income is weaker. While income levels tend to be higher in Superstar cities, the results are not robust across specifications, and there is little evidence of systematic associations between demand and supply dynamics and local incomes.

The thesis is structured as follows. Chapter 2 reviews the relevant literature, focusing on drivers of housing demand and supply, as well as a short description of recent work about Germany. Chapter 3 presents a simple model of household location choice based on income and preferences for cities, outlines the empirical strategy, and describes the data. Chapter 4 reports and discusses the empirical results. Finally, Chapter 5 concludes.

2. Literature Review

Across the European Union, there has an increasing share of the population moving into cities. Germany is no exception to such trend, particularly among the younger generations who disproportionately opt for urban regions as places of residence.[1] The advantages of living within denser urban regions come from concentrated economic activities, which facilitate knowledge spillovers and generate productivity gains through agglomeration forces such as labor market pooling, firm specialization, and learning externalities.

At the same time, the growing concentration of population and economic activity in cities has intensified pressures on housing markets. Many urban areas face increasingly binding housing supply constraints, rising rents, and growing spatial inequalities. Such developments raise questions about who benefits from urban growth and whether access to productive labor markets is becoming restricted solely to those who can afford to outbid others for the same location. These challenges are not limited to a specific region: evidence from both developed and developing economies shows that housing affordability has become a central issue in high-productivity cities around the world (Glaeser, 2012). Consequently, policymakers search for solutions through zoning reform, place-based policies, and housing subsidies, such as rent freezes, as tools to promote more equally distributed growth without losing consensus. The central question raised in the literature is the following: to what extent do housing supply constraints interact with rising urban demand to generate persistent differences in prices and incomes across cities?

2.1. Housing Markets and Urban Constraints

A central insight from the urban economics literature is that high house prices originate from the interaction between demand for location and the elasticity of housing supply. On the demand side, a large body of literature emphasizes the role of agglomeration forces (Duranton and Puga, 2004; Krugman, 1992; Marshall, 1890; Rosenthal and Strange, 2001). Cities facilitate knowledge spillovers, labor market pooling, and firm specialization, all of which contribute to higher productivity and wage levels. As a result, workers and firms are drawn to denser urban areas not only because of higher earnings but also due to the concentration of opportunities. In some cases, rising house prices may therefore reflect not only higher productivity, but also an increasing valuation of denser urban space, which further amplifies demand pressures.

Yet, the degree to which higher demand translates into higher prices depends critically on the responsiveness of housing supply. When supply is elastic, new construction can accommodate increases in demand, limiting upward pressure on prices. By contrast, when supply is inelastic, demand shocks are primarily capitalized into higher housing costs (Duranton and Puga, 2023; Glaeser and Gyourko, 2018). Logically, if urban developers manage to maintain the construction pace, or renovation, of new housing units, then there should be fewer affordability issues since the residential stock would allow for the necessary

[1] https://ec.europa.eu/eurostat/statistics-explained/index.php?title=Urban-rural$_{E}urope_{-d}emographic_{d}evelopments_{i}n_{r}ural_{r}egions_{a}nd_{a}reasHighlights$ This "Statistics Explained" section of Eurostat provides a snapshot of the urban-rural divide within the European Union.

space incoming residents. There are many reasons why this seems not to be the case. An example is geographic constraints. Saiz (2010) provides evidence for topological driven limits to development, including large bodies of water or steeper terrain slopes as sources of reduced supply elasticities. In Gyourko et al. (2013), there also an attempt to use these proxies for an ex-ante definition of supply elasticity and they too find a significant association between this definition of supply and the emergence of superstars. In addition, land use regulations, zoning restrictions, and building density limits can further constrain new construction, even in the absence of physical barriers.

The interaction of strong demand and constrained supply gives rise to the phenomenon of "Superstar cities" as formalized by Gyourko et al. (2013). Through very simple assumptions of location preference and individual productivity, they model cities characterized by high demand and low supply. They show that these cities tend to attract higher-income households, leading to a rightward shift in the income distribution relative to the US median.

2.2. The German Context

The relevance of housing market dynamics depends on country specific institutional and market characteristics. In the German context, several studies provide evidence of both demand driven price increases and limited supply responsiveness.

Lerbs (2014) examines housing supply using panel data fro 401 German counties and cities over the period from 2004 to 2009, focusing specifically on new single-family construction. Their paper finds that higher house prices increase housing supply, although the response is weak. Interestingly, rising development costs are associated with more construction, pointing to increasing costs as an indicator of strong demand or rising expectations, rather than development constraints alone. Overall, German housing supply reacts less strongly to price changes than in countries like the U.S.

Brausewetter et al. (2022) analyze housing prices across districts from 2008 to 2019 using a broader dataset covering all housing types. Their results show large persistent regional disparities, with prices increasing mostly in larger cities and southern regions that were already expensive. They highlight demand drivers of such rising costs such as population changes and higher incomes. The share of highly educated workers rises and as incomes follow, then prices increase too, thus pointing to a demand driven price surge. In general, the underlying argument is that cities attract talent, more specialized jobs, and higher incomes moving for better amenities. Therefore, given the slow and constrained housing supply, prices adjust upward to clear the market.

Beze (2023) focuses on estimating housing supply elasticity and the role of geographic constraints (e.g., water bodies or steep slopes) using a 2SLS approach with a Bartik instrument to identify exogenous demand shocks (Bartik, 1991). Their objective is to test whether physical constraints due to geographic characteristics of the land limit new development, as in Saiz (2010). The paper finds that between 2008 and 2019, housing supply is strongly inelastic, with elasticity estimates of about 0.22 considering floorspace and 0.25 for housing units. This implies that a 10% increase in prices leads to only about 2.2% increase in housing supply. Contrary to expectations, geographic constraints do not seem

to be a key bottleneck for new development. Instead, the main constraint is argued to be building intensity: areas with a high share of already developed land, especially within urban districts, exhibit significantly lower supply responsiveness. Surprisingly, they also find little difference in price growth between West and East German districts, while the main gap seems to be between urban and rural, reflecting stronger demand pressures and tighter supply constraints in cities.

2.3. Relevance and Literature Gap

Building on early insights from Marshall (1890), a large body of literature has sought to explain why economic activity is concentrated in cities and how this concentration affects wages and productivity. Empirical evidence shows that workers in denser labor markets experience higher wage growth and productivity gains. For instance, Glaeser and Maré (2001) document faster wage growth in cities, while Moretti (2004) find that a higher share of educated workers raises wages for all workers in a given location, supporting the hypothesis that, conditional on density, human capital spillovers positively affect wage regardless of the income level. Depending on industry structure, firms also face significant advantages in locating within denser areas. Overman and Puga (2010) suggest that firms prefer larger worker pools to "iron out" idiosyncratic shocks. By locating near a dense concentration of workers, firms improve their ability to shelter against bad times.

At the same time, the literature emphasizes that both worker sorting and agglomeration effects contribute to spatial wage differences. Combes et al. (2008) show that more productive workers tend to sort into cities, but that agglomeration itself has an independent effect on wages. These dynamics imply that cities not only attract higher skilled individuals (assuming income as a proxy for skill), but also enhance productivity through local interactions. However, the benefits of agglomerations are not necessarily evenly distributed. Eeckhout et al. (2014) argue that sorting by skill across cities increases wage dispersion, suggesting that urbanization may increase inequality alongside average income.

Another relevant aspect is the role of housing markets in mediating these effects. In particular, if access to cities with high productivity is limited by housing supply constraints, then the spatial allocation of workers may be distorted. For example, housing constraints have implications for migration and labor allocation. Hsieh and Moretti (2019) argue that restrictive housing supply in high productivity cities limits worker migration, leading to aggregate productivity losses due to spatially misallocated labor. Similarly, Ganong and Shoag (2017) provide evidence that rising housing costs in productive regions have contributed to a decrease in internal migration within the United States.

This paper contributes to the literature by testing the predictions of the Superstar cities model in the German context. By combining detailed data on housing prices, construction, and income across cities over a relatively more extended period, it evaluates whether locations affected by high demand and supply limitations exhibit both higher housing prices and patterns of income consistent with spatial equilibrium models and sorting mechanisms. The analysis provides new evidence on the extent to which housing market frictions shape not only price dynamics, but also the allocation of households and the distribution of income across space.

3. Model, Data and Methodology

3.1. Model

The two locations framework from Gyourko et al. (2013) is based on a few simple assumptions. The model considers two cities that differ in the elasticity of housing supply. In the first location, housing supply is perfectly elastic and additional units can always be constructed so that rents are normalized to zero. In the second location, housing supply is constrained. The housing stock expands only if rents increase sufficiently to make new construction profitable, which implies that supply responds sluggishly to demand shocks.

Household utility is increasing in both location preferences and consumption, where consumption depends on the income earned in each location. Households differ in their valuation of locations, which captures heterogeneity in preferences for local amenities such as climate, public goods, or non-housing consumption opportunities. The distribution of preferences is such that, in the absence of housing costs, demand would be sufficient to fully occupy the constrained city. Productivity determines wages, which are assumed to reflect marginal productivity. Importantly, preferences and productivity are assumed to be independently distributed, so households with different income levels are equally likely to value either location. Finally, households maximize utility derived from taste for the city they live in and non-housing consumption. Because housing supply is constrained in the inelastic city, rents must adjust to equate demand with fixed housing stock and clear the market. If this were not the case, then there will always be at least one household willing to move to increase their utility. As rents increase, only those with sufficiently strong preferences or sufficiently high productivity will be willing to live in the inelastic city. Thus, each agent faces a trade-off between rent and consumption that depends on taste and income.

Formally, household wage w and location preference c are independent of each other and are defined on $[0, \infty)$ and $[0, 1]$ respectively. Let c denote the preference for the inelastic city, such that $c = 1$ corresponds to the household with preference for the inelastic location above every other place. The household's utility from locating in the elastic city E is $U^E = (1 - c)w$, whereas utility in the inelastic city I is $U^I = c(\beta w + \alpha - r)$, with $\beta w + \alpha$ representing the wage she would earn in city I and r being rent costs. Focusing on the case in which demand for housing in city I is larger than the amount of units supplied, households face the following maximization problem: $U = \max\{U^E, U^I\}$.

Utility maximizing households will prefer to locate in city I when $U^I \geq U^E$. Substituting in this condition and rearranging terms we find that:

$$c_c(w) = \frac{w}{(1 + \beta)w + \alpha - r}$$

The expression defines the minimum level of preference required, for a given wage, to choose the inelastic city. In other words, conditional on income, households will locate in I only if their preference is larger than a certain threshold $c_c(w)$. As rents increase, the threshold $c_c(w)$ rises, implying that only households with stronger preferences for the

constrained city will choose to locate there.

On the other hand, we find a threshold for income:

$$w_c(c) = \frac{(r-\alpha)c}{(1+\beta)c-1}$$

That is, given location preferences and productivity differences from choosing between the two cities, a household will prefer to live in I if income is larger than a cutoff value w_c. Higher rents also increase the income threshold $w_c(c)$, reinforcing the selection of higher-income households.

The model yields two central predictions. First, higher rents and, under certain conditions, higher average wages due to sorting. Second, the income distribution in the constrained city will shift toward higher-income households relative to the elastic city. Together, these two mechanisms provide a theoretical foundation for the emergence of so-called "Superstar" cities characterized by persistent price growth and income sorting.

These theoretical predictions guide the empirical analysis. In particular, the model implies that cities characterized by more inelastic housing supply should exhibit higher housing prices and, potentially, differences in income levels and distribution due to sorting. The empirical strategy therefore focuses on identifying whether such patterns can be observed in the German context.

3.2. Data

The main data sources for the analysis are the German Real Estate Price Index (GREIX) database and the *Statistische Ämter des Bundes und der Länder* (regional offices), for house values and construction completions respectively. Data on income per capita at city level is taken from Eurostat datasets.

The GREIX database, developed in Amaral et al. (2023), exploits mandatory registration of housing purchases with the *Gutachterausschuss*, which in turn stores records and compiles statistics on transaction volumes and price changes. For Western Germany, such practice has been mandatory since the 1960s. They further construct quality-adjusted price indices using rolling window time-dummy hedonic regressions for each city. Transaction prices are regressed on property characteristics (e. g. lot size, number of rooms, construction year, neighborhood) and year dummies, thus allowing the value the market assigns to each characteristic to change over time.

Regional statistical offices collect information on both the flow of dwelling permits as well as building completions. Because permits are granted before the start of construction, they are typically interpreted as a leading indicator of building activity and local economic conditions, often several quarters in advance. However, not all permitted projects are ultimately realized, and the timing between authorization and delivery can vary substantially across locations and market phases. Building completions, in contrast, capture the actual addition to the housing stock and therefore provide a direct measure of realized supply. Nevertheless, permits and completions tend to move closely together at the aggregate level; for example, Lerbs (2014) reports correlations of roughly 0.95 for Germany over the period 1995-2009. For the purposes of my thesis, I use construction completions. A third

fundamental component of the panel consists of income data. To test whether the model distribution prediction holds, i. e. only higher income households will select into the supply constrained city, I use data on the median and the average disposable annual household income from Eurostat for each city in the panel.[1]

This leaves a dataset on income, house values, and new dwellings from sixteen *Kreisfreie Städte* (governing districts for cities that are large enough in terms of income and population to self-administer themselves) and two federal states, Berlin and Hamburg. The panel covers the period from 1995 to 2024 and totals 108 data points, one for each city-period observation.

3.3. Empirical Strategy

To reduce noise in the data, the panel is divided into five 5-year periods.[2] For each unit i and year t, growth is measured as the constant yearly rate that would take the value of the variable from its level in year $t-5$ to its level in year t. Formally, the rate $g_{i,t}$ is computed as

$$g_{i,t} = \left(\frac{X_t}{X_{t-5}}\right)^{\frac{1}{5}} - 1 \tag{3.1}$$

where $\frac{X_t}{X_{t-5}}$ denotes the ratio between the price level or the number of dwellings in year t and their 5-year lag. This transformation converts the total change observed over five years into an equivalent average annual growth rate, making growth comparable across time periods and between units.

Table 3.1 presents summary statistics for each city in the sample. The rows highlighted in light blue indicate cities that were part of the former GDR. Columns 2 to 5 report statistics for price growth. For reference, the average period price growth of the overall index for Germany amounts to 1.2%. Of the 18 cities in the sample, 11 show an average period price growth above the national level. Nevertheless, there is still a great degree of heterogeneity, with cities such as Potsdam or Munich doubling this number, whereas other places like Chemnitz and Duisburg display negative average period growth.

Annualized growth in dwellings show an even more pronounced discrepancy between cities that expand their housing stock extensively and cities with limited increases construction activity. Unsurprisingly, Duisburg and Chemnitz display the lowest average yearly increase in new housing. At the same time, other wealthier cities exhibit relatively modest housing stock expansion. For example, Frankfurt (0.74%) and Hamburg (0.64%) show a lower growth in housing units than the sample median of 0.77%. Finally, particularly notable are the levels of growth and persistence displayed by cities like Münster and Potsdam, where the average annualized 5-year growth in housing units is 4.25% and 2.5% respectively with minimum values that are roughly twice the sample median.

[1] Eurostat data from the urb_clivcon dataset as part of the Urban Audit and the Large City Audit project. Collected on 17/02/2026.

[2] The choice of five-year periods is partly pragmatic. Given the length of the panel and the limited number of cities, this aggregation balances the need to keep a sufficient number of observations while smoothing short run fluctuations and capturing medium run housing market dynamics.

Table 3.1.: Annualized 5-Year Growth Statistics (%)

	Price Growth (Apartments)				Dwellings Growth			
City	Mean	SD	Min	Max	Mean	SD	Min	Max
Berlin	2.22	6.21	-4.34	10.65	0.77	0.53	0.22	1.43
Bonn	1.39	4.20	-2.62	8.74	0.63	0.24	0.44	1.07
Chemnitz	-1.02	5.18	-5.29	8.47	0.40	0.46	0.15	1.34
Dortmund	0.54	4.33	-4.26	7.80	0.45	0.16	0.29	0.69
Dresden	0.84	4.35	-3.79	7.60	0.75	0.78	0.19	2.28
Duisburg	-0.34	4.74	-5.88	6.75	0.30	0.17	0.17	0.61
Düsseldorf	1.61	4.45	-2.33	8.40	2.87	1.92	1.62	6.55
Erfurt	1.48	3.98	-3.46	6.58	0.51	0.38	0.25	1.22
Frankfurt	1.64	5.11	-3.04	9.44	0.74	0.16	0.56	1.01
Hamburg	1.92	4.29	-1.88	7.76	0.64	0.22	0.37	0.86
Karlsruhe	1.14	4.76	-4.13	8.26	0.78	0.34	0.56	1.47
Köln	1.77	4.03	-2.03	7.98	2.44	1.41	1.60	5.19
Lübeck	0.95	5.23	-4.80	8.91	0.52	0.25	0.35	0.98
München	2.34	5.50	-2.46	10.13	0.98	0.09	0.84	1.06
Münster	1.79	4.17	-2.64	7.34	4.25	3.00	2.52	10.09
Potsdam	2.40	5.31	-6.88	8.21	2.52	1.92	1.38	6.37
Stuttgart	1.38	4.99	-3.85	8.49	0.82	0.33	0.61	1.47
Wiesbaden	0.47	4.39	-3.45	7.52	0.51	0.22	0.33	0.80

3.4. Defining Superstar

A city is classified as a Superstar if it simultaneously exhibits above-median demand pressure and above-median inelasticity of supply.[3] Demand pressure in a city is defined as the sum of price growth and housing unit growth, capturing the overall intensity of market expansion. The idea is that if a city builds many new units while prices remain stable, it is still considered a high demand location. Supply inelasticity of housing with respect to price is measured as the ratio of housing price growth over unit growth, reflecting the extent to which prices respond to new construction. A larger ratio indicates that prices respond strongly while construction responds weakly, suggesting more inelastic housing supply.

This definition ensures that Superstar cities are characterized by both strong demand and relatively constrained housing supply. Gyourko et al. (2013) identify Superstars in period t if the cutoff conditions are satisfied in the two previous periods. However, from their replication package the supply proxy is defined in a "Static" manner over the entire sample. That is, a city is supply constrained if its supply inelasticity in 2000-2004 exceeds the cutoff across the entire sample.[4] In my analysis, I add a "Lagged" definition in which the cutoff values are determined using the two previous periods. For example, a city is

[3] The original cutoff value from Gyourko et al. (2013) is for a city to be in the top decile of inelasticity of supply. Given the limits of the sample for Germany, a more generous threshold, i. e. the median, has been adopted.

[4] Gyourko, Joseph, Mayer, Christopher, and Sinai, Todd. Replication data for: Superstar Cities. Nashville, TN: American Economic Association [publisher], 2013. Ann Arbor, MI: Inter-university Consortium for Political and Social Research [distributor], 2019-10-13. https://doi.org/10.3886/E114839V1

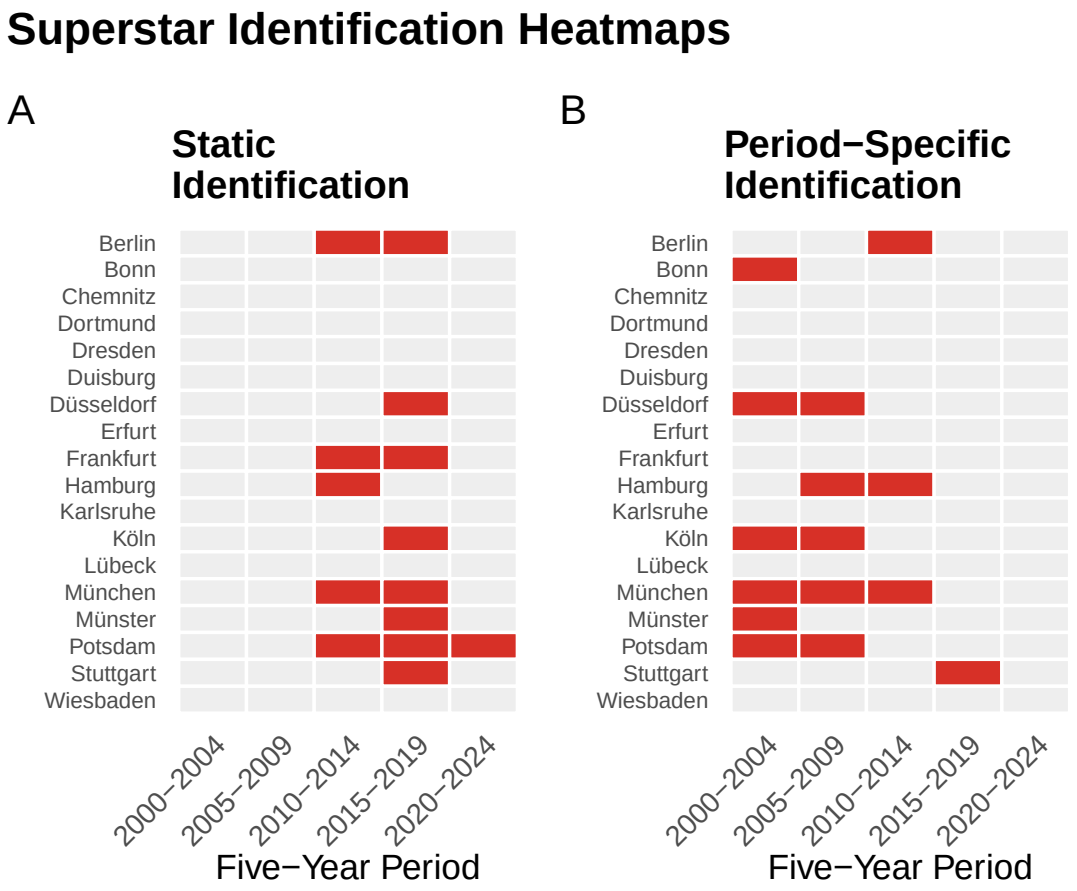

Figure 3.1.: Classification of German cities as Superstar housing markets across five-year periods from 2000 to 2024. Panel A applies the static identification strategy, where inelasticity thresholds are compared with the sample median. Panel B applies lagged cutoff identification strategy, where thresholds are computed using two-period windows. Cities marked in red are classified as Superstars. The training window 1995-1999 is excluded from the visualization. For a visualization without the COVID period (2020-2024), see Appendix A.

labeled as Superstar in the period 2000-2004 if its values of high demand and inelastic supply are above their respective mean in periods 1995-1999 and 2000-2004.

Figure 3.1 summarizes the resulting Superstar classification.[5] City-periods are represented with rectangles and market types are defined by colors. Red rectangles indicate that a city is defined as Superstar, while gray rectangles indicate non-Superstar cities. With both identification strategies, one can find similarities. There are a group of cities which never become superstars throughout the sample. These include former GDR cities such as Dresden and Chemnitz, as well as other cities like Dortmund or Duisburg of the former FRG that experienced lower growth since the 1990s.

Two mechanical differences between identification strategies emerge. Under the static

[5]The five-year period 2020-2024 has been characterized by a significant fall in house values and very limited new construction rate. Naturally, if one uses the static definition of supply, thus considering the entire sample for the median cutoff, such higher volatility in prices and units growth may significantly affect the results. A heatmap excluding the COVID-19 pandemic is provided in Appendix A. Data from the GREIX database, apartment purchases.

cutoff identification, Superstars are concentrated in the most recent section of the sample, whereas in the lagged identification they are mostly concentrated in the 2000s. In the former case, this pattern arises because inelasticity of supply is measured over the whole sample. The higher price growth experienced in the last three periods increases the cutoff values and thus only during more recent years the numerator of the supply ratio is going to be large enough for the fraction to pass the cutoff. In the latter, cities which are historically in high demand are identified as Superstars at earlier stages. Over time, however, as more productive cities become unaffordable, households' location decisions will also include more peripheral places. As demand spreads geographically, price growth will affect more locations. This idea can be empirically observed through the static cutoff panel. At first, in period 2010-2014 only five of the city-period rectangles are identified as Superstars. As one moves onto the next period (2015-2019) and different cities become more competitive, the number of Superstars increases to eight out of the eighteen total. In addition, within a relatively small sample (built mostly from large regional centers), this diffusion of demand reduces the relative distinction between cities, so that no single location persistently dominates others in terms of density, wage premia, or amenities. A larger and more heterogeneous sample might reveal larger differences across cities and thus provide clearer support for the Superstar mechanism.

An example of this spatial reallocation emerges during the COVID-19 pandemic. Under the static cutoff identification, only Potsdam retains Superstar status. This likely reflects its geographic proximity to Berlin combined with the expansion of remote work. Easier commuting into Berlin and greater availability of larger home space, drastically increased demand for housing relative to more densely built urban cores. Similar patterns have been documented in the United States, where remote work significantly contributed to post-pandemic house price growth (Mondragon and Wieland, 2022).

3.5. Regression Specifications

To test the implications of the model, two regressions are estimated. The first specification uses a fixed Superstar indicator for the entire panel, while the second allows Superstar status to vary over time and includes city fixed effects.

$$\log(Y_{it}) = \beta_1 \, \text{Superstar}_i + \delta_t + \varepsilon_{it} \tag{3.1}$$

$$\log(Y_{it}) = \beta_1 \, \text{Superstar}_{it} + \delta_t + \gamma_i + \varepsilon_{it} \tag{3.2}$$

In both cases, log house prices and log local income Y_{it} are regressed on the Superstar indicator. In addition, period fixed effects δ_t are included to control for nationwide macroeconomic shocks that simultaneously affect all cities, such as changes in interest rates, credit conditions, or aggregate demand. For example, a monetary tightening or expansion by the European Central Bank would equally affect the entire country. Including period controls ensures that the estimated effect of Superstar status is identified from cross-city variation within the same period, rather than being driven by common time trends in housing prices.

The second specification 3.2 also includes city fixed effects γ_i to control for time invariant unobserved heterogeneity across cities that could confound the results. These fixed effects

can be interpreted as absorbing the set of local characteristics that hardly change over time, including amenities (e.g. climate, crime, long-run quality of life) and persistent differences in productivity. By including γ_i the regression is differencing out the cross sectional variation between cities, so identification of β_1 comes only form changes over time within a city. It follows that the estimate of β_1 captures the association between house values and income levels and a city transitioning into Superstar status, while holding constant any fixed local attributes.

Finally, since nearby cities may be spatially correlated, standard errors are clustered at city level to address heteroskedasticity and avoid overstating the statistical power of the OLS estimator. More information on the unbiasedness of the OLS estimator and heteroskedastic errors is provided in Appendix B.

4. Results

This section analyzes the results from the regression specifications outlined in the previous section. The objective is to test the model's main propositions: namely, conditional on demand, house values and income levels are higher in cities with greater inelastic supply of housing.[1] To assess the former proposition, I use data on apartment purchases and single family houses in Germany. For the latter, I rely both average as well as median disposable income per household at the city level.

The tables in this section include four alternative specifications. These are distinguished mainly by the identification strategy of the inverse supply elasticity i. e., whether the supply cutoff is defined using a static or lagged measure as described in the previous section. Second, the results report coefficient from both the pooled-cross sectional specification, with period fixed effects, and the time-varying specification that exploit within-city variation, accounting for local persistent characteristics. As far as the data allow, all regressions include the COVID period from 2020 to 2024.

The analysis of income distributions begins with the construction of the city-level income dataset. Data are divided into two categories based on the income measure: median disposable household income and average disposable household income. These subsets are labeled accordingly and recombined into a single dataset for consistent processing. The data are then transformed into a balanced panel covering the period 2000–2019. Since income data for the period 2020-2024 are missing, the total number of data points available for the analysis on income distribution is 90. During this step, missing years are explicitly interpolated for small gaps in the income series (up to three consecutive years) for each city-period combination. The national median income is also adjusted to respect the baseline of the original dataset.[2]

A key prediction of the model is that the income distribution in Superstar cities is shifted to the right relative to the national distribution. The proxy used in this analysis is the difference between a city's median income and the national median. A larger positive difference implies that the city has substantially higher wages for a larger share of the population, compared to the national median. Such findings would be consistent with a shift to the right of the wage distribution. For specification 3.1, the empirical implementation is

$$\log\left(\frac{\text{Median Income}_{it}}{\text{Median National Income}_t}\right) = \beta\,\text{Superstar}_i + \delta_t + \varepsilon_{it}$$

Where the coefficient β captures the percentage difference between Superstars and other cities in terms of the distance from mean to median.

Finally, within-city distance between mean income and median income provides some, though limited, indication of how income is distributed. Including this outcome variable

[1] In this section, the terms house values and house prices are used interchangeably.

[2] Data for the manual correction from Eurostat database on Income and Living Conditions (ilc_di03) retrieved on 22/03/2026. National statistics on median income up to 2012 are from the urb_clivcon database mentioned in Chapter 3. From 2013 to 2018 data is missing; therefore, to maintain a balanced panel, growth rates from ilc_di03 are applied to fill the gap.

is a way to assess whether changes associated with the emergence of Superstar cities are accompanied by shifts in income dispersion. With income data, it is usually the case that the mean is larger than the median, suggesting a right skewed distribution in which a relatively small number of high-income individuals pull the average upward. Tracking how this gap evolves alongside the Superstar classification can indicate whether household location decisions due to housing consumption coincide with changes in the top earners of the distribution. Empirically,

$$\log\left(\frac{\text{Mean Income}_{it}}{\text{Median income}_{it}}\right) = \beta\,\text{Superstar}_i + \delta_t + \varepsilon_{it}$$

Where β captures the difference in the mean-to-median income ratio between Superstar and non-Superstar cities, for each city i and period t.

4.1. Apartment Data

4.1.1. Price Dynamics

Table 4.1 presents the results for apartment prices across cities. Using the static identification, reported in columns 1 and 2, cities classified as Superstars exhibit significantly higher house values. The coefficient for the fixed Superstar indicator is positive and statistically significant, indicating that Superstar cities are associated with 27% higher apartment prices when controlling for period effects. When Superstar status is allowed to vary over time, the estimated effect remains positive and significant, although somewhat smaller, suggesting that periods in which a city meets the Superstar criteria are associated with an increase of roughly 19% in apartment prices.

In contrast, results for the lagged identification strategy in columns 3 and 4 yield weaker results. While the fixed Superstar indicator remains positive and statistically significant, the time-varying specification becomes statistically insignificant once city fixed effects are included. The findings suggest that when supply inelasticity is defined using a rolling window based on past observations, Superstar cities exhibit higher house prices than other locations. However, once city fixed effects are included, the estimated relationship cannot be distinguished from unobserved, time-invariant local characteristics. For example, cities like Munich or Berlin may have higher prices for houses relative to other places, but this could be entirely due to long standing advantages deriving from higher productivity, desirable amenities, or any factor that is especially persistent over time and is absorbed by the city fixed effects.

The results consistently indicate that Superstar cities exhibit higher housing prices. However, once city fixed effects are introduced, much of this variation is absorbed, suggesting that the Superstar classification may be capturing persistent, time-invariant city characteristics rather than purely dynamic effects. Under the lagged identification a city's classification depends on its position relative to other cities at a given point in time. Consequently, status is sensitive to changes in the cross-sectional distribution (over the previous two periods): even if a city's fundamentals remain unaltered, it may change classification as the benchmark cutoff shifts. With a small sample of 18 cities, where many places experience similar trends in prices and construction, this moving cutoff can limit cross-city

Table 4.1: City Differences in Prices (Apartments)

	Log House Value Pooled-Cross Section	Log House Value Time-Varying	Log House Value Pooled-Cross Section	Log House Value Time-Varying
Superstar$_i$	0.271***		0.245***	
	(0.054)		(0.051)	
Superstar$_{it}$		0.193***		-0.059
		(0.052)		(0.049)
Constant	4.420***	4.564***	4.413***	4.632***
	(0.052)	(0.064)	(0.053)	(0.070)
Num.Obs.	90	90	90	90
RMSE	0.21	0.11	0.21	0.12
Std.Errors	Clustered	Clustered	Clustered	Clustered
Identification	Static	Static	Lagged	Lagged
Fixed effects	Period	Period + City	Period	Period + City

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

differences. This may result in too little identifying variation to capture any significant difference.

4.1.2. Income Dynamics

Table 4.2 shows the relationship between Superstar status and various measures of local income, including average income levels, deviations from the national median, and income dispersion within the city. Across all specifications, there is no robust evidence that Superstar cities exhibit systematically higher income levels or stronger income growth. This contrasts with the model's prediction of income sorting and suggests that housing market outcomes are only weakly connected to observable income dynamics.

In Panel A, the coefficients are generally small and statistically insignificant. This applies both to the static and lagged identification strategies, including specifications with city fixed effects. A similar pattern emerges in Panel B, where the distance from the national median income does not show a consistent or significant association with Superstar classification. Panel C also provides no statistically significant relationship between Superstar status and internal income dispersion. Although the coefficients are generally positive, they are imprecisely estimated and become even smaller once within-city variation is exploited. Only measures of Superstar status based on the lagged identification strategy of housing supply inelasticity exhibit robust correlations with all income outcomes.

The results together suggest that the higher housing prices observed in Superstar cities are not systematically associated with higher income levels or different income dynamics compared to the rest. While some coefficients are positive, none are statistically distinguishable from zero. Instead, results are probably driven by persistent structural characteristics such as amenities, long standing demand, or supply constraints, rather than observable differences in local income dynamics. Consequently, city fixed effects likely absorb much of the relevant cross-sectional variation associated with these persistent characteristics.

Table 4.2: Income Differences Across Cities (Apartments)

Panel A: Log Mean Income Per Capita				
	Pooled-Cross Section	Time-Varying	Pooled-Cross Section	Time-Varying
Superstar_i	0.046		0.056	
	(0.064)		(0.066)	
Superstar_{it}		0.014		0.005
		(0.016)		(0.008)
Constant	10.042***	9.988***	10.033***	9.991***
	(0.047)	(0.007)	(0.053)	(0.007)
Num.Obs.	72	72	72	72
RMSE	0.13	0.03	0.13	0.03
Std.Errors	Clustered	Clustered	Clustered	Clustered
Fixed effects	Period	Period + City	Period	Period + City
Identification	Static	Static	Lagged	Lagged
Panel B: Log Distance from National Median				
Superstar_i	0.024		0.033	
	(0.051)		(0.052)	
Superstar_{it}		0.010		0.005
		(0.015)		(0.010)
Constant	−0.014	−0.092***	−0.021	−0.090***
	(0.034)	(0.010)	(0.038)	(0.009)
Num.Obs.	72	72	72	72
RMSE	0.11	0.03	0.11	0.03
Std.Errors	Clustered	Clustered	Clustered	Clustered
Fixed effects	Period	Period + City	Period	Period + City
Identification	Static	Static	Lagged	Lagged
Panel C: Internal Income Dispersion				
	Pooled-Cross Section	Time-Varying	Pooled-Cross Section	Time-Varying
Superstar_i	0.023		0.024	
	(0.016)		(0.016)	
Superstar_{it}		0.004		−0.001
		(0.012)		(0.009)
Constant	0.199***	0.222***	0.197***	0.224***
	(0.015)	(0.007)	(0.017)	(0.006)
Num.Obs.	72	72	72	72
RMSE	0.04	0.02	0.04	0.02
Std.Errors	Clustered	Clustered	Clustered	Clustered
Fixed effects	Period	Period + City	Period	Period + City
Identification	Static	Static	Lagged	Lagged

+ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

4.2. Single-Family House Data

In order to have a broader perspective of the Superstar phenomenon, this section will examine the effect of the interaction between high demand and limited supply for single-family dwellings. In 2024, single-family house construction completions collectively comprised almost 73% of total new building completions while representing only 25% of total new dwellings.[3] This indicates that single-family housing remains a major incentive for developers of new construction in Germany despite accommodating a smaller share of households. In the GREIX database, price data for single-family housing in Munich are not available for the period 1995 to 2019. As a result, Munich is excluded altogether from the analysis, reducing the number of observations from 90 to 85. Finally, data for years after 2020 are not available for the income section, again reducing the number of observations to 68.

4.2.1. Price Dynamics

The regression results for single-family house prices are reported in Table 4.3. Compared to the apartment results, the overall pattern for the estimated coefficients remains similar, although the magnitudes are somewhat smaller. Under the static identification (columns 1 and 2), cities that are ever classified as Superstar exhibit higher house values. The pooled cross-sectional specification indicates house prices approximately 15% higher in Superstar cities, while the time-varying specification suggests that periods in which a city attains Superstar status are associated with an increase of roughly 17% in prices.

Under the lagged identification strategy (columns 3 and 4), the pooled cross-sectional specification continues to show a positive and statistically significant association, with Superstar cities exhibiting house prices around 22% higher than other cities. However, once city fixed effects are included, the coefficient on the time-varying Superstar indicator becomes small and statistically insignificant. Similarly to the apartment results, this suggests that much of the observed price premium reflects persistent differences across cities rather than dynamic within-city changes over time.

4.2.2. Income Dynamics

Overall, the results again provide limited and inconsistent evidence of a systematic link between Superstar classification and income outcomes. In Panel A of Table 4.4, which examines income per capita, neither the static nor the lagged identification strategies produce statistically significant coefficients, both across and within cities. A similar absence of robust relationships appears in Panel B for the distance from the national median income.

Panel C, which examines internal income dispersion, also provides little evidence of systematic differences between Superstar and non-Superstar cities. Although the lagged pooled cross-sectional specification yields a positive coefficient of approximately 0.031, the estimate is statistically insignificant and disappears once city fixed effects are included. Overall, the results suggest that Superstar classification in the single-family housing market is not strongly associated with higher income levels or greater income inequality. Instead, similarly to the apartment results, the observed housing price differences are more likely

[3] https://www-genesis.destatis.de/datenbank/online/statistic/31121/table/31121-0009/chart/pie retrieved on 18/03/2026

Table 4.3: City Differences in Prices (Single-Family Houses)

	Log House Value Pooled-Cross Section	Log House Value Time-Varying	Log House Value Pooled-Cross Section	Log House Value Time-Varying
Superstar$_i$	0.147*		0.217**	
	(0.065)		(0.077)	
Superstar$_{it}$		0.165***		0.045
		(0.039)		(0.030)
Constant	4.395***	4.472***	4.316***	4.475***
	(0.053)	(0.026)	(0.065)	(0.032)
Num.Obs.	85	85	85	85
R2 Adj.	0.674	0.888	0.730	0.873
Std.Errors	Clustered	Clustered	Clustered	Clustered
Fixed effects	Period	Period + City	Period	Period + City
Identification	Static	Static	Lagged	Lagged

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

driven by persistent structural and locational characteristics rather than by observable income sorting dynamics.

These results are somewhat noteworthy given the different economic role typically associated with single-family housing relative to apartments. Single-family homes are generally more closely linked to higher-income households, suburban development, and owner-occupancy, while apartments tend to accommodate a broader range of households and are more strongly associated with dense urban environments. One might therefore expect income sorting to appear more clearly in the single-family segment, particularly if wealthier households disproportionately concentrate in constrained, high-demand cities. However, the results provide little evidence of stronger income dynamics even in this market segment.

4.3. Discussion

These findings suggest that, in Germany, the Superstar phenomenon operates primarily through housing market dynamics rather than income sorting. An interpretation is that housing prices in constrained cities reflect not only income gaps but also persistent and unobserved advantages specific to certain locations. This means that diverging house prices seem to reflect differences in willingness to pay for a specific location more than differences in observed income levels. Even in the absence of significantly higher average incomes, households may still bid up prices in certain cities because of better amenities, stronger access to the labor market, and other benefits that are not strictly monetary.

This interpretation is consistent with the Rosen-Roback spatial equilibrium model (Roback, 1982; Rosen, 1979), which provides a framework to understand how different variables other than income and housing consumption play a role in households' utility maximizing problem. In this setting, individuals trade off wages, housing costs, and local amenities when choosing a location. that, in Germany, the Superstar phenomenon operates primarily through housing market dynamics rather than income sorting. While the Rosen-Roback model is primarily a framework to evaluate the role of local amenities, it also offers insights for interpreting these results by highlighting how non monetary features

Table 4.4: Income Differences Across Cities (Single-Family Houses)

Panel A: Log Mean Income Per Capita				
	Pooled-Cross Section	Time-Varying	Pooled-Cross Section	Time-Varying
Superstar_i	−0.051		0.044	
	(0.064)		(0.079)	
Superstar_{it}		0.036		0.000
		(0.024)		(0.016)
Constant	10.088***	9.990***	10.024***	9.995***
	(0.051)	(0.006)	(0.069)	(0.007)
Num.Obs.	68	68	68	68
RMSE	0.12	0.03	0.12	0.03
Std.Errors	Clustered	Clustered	Clustered	Clustered
Fixed effects	Period	Period + City	Period	Period + City
Identification	Static	Static	Lagged	Lagged
Panel B: Log Distance from National Median				
	Pooled-Cross Section	Time-Varying	Pooled-Cross Section	Time-Varying
Superstar_i	−0.049		0.013	
	(0.048)		(0.061)	
Superstar_{it}		0.023		−0.006
		(0.027)		(0.016)
Constant	0.025	−0.083***	−0.013	−0.078***
	(0.033)	(0.010)	(0.051)	(0.009)
Num.Obs.	72	72	72	72
RMSE	0.10	0.03	0.10	0.03
Std.Errors	Clustered	Clustered	Clustered	Clustered
Fixed effects	Period	Period + City	Period	Period + City
Identification	Static	Static	Lagged	Lagged
Panel C: Internal Income Dispersion				
	Pooled-Cross Section	Time-Varying	Pooled-Cross Section	Time-Varying
Superstar_i	−0.003		0.031	
	(0.019)		(0.020)	
Superstar_{it}		0.012		0.005
		(0.016)		(0.006)
Constant	0.210***	0.221***	0.185***	0.221***
	(0.021)	(0.007)	(0.020)	(0.007)
Num.Obs.	68	68	68	68
RMSE	0.04	0.02	0.03	0.02
Std.Errors	Clustered	Clustered	Clustered	Clustered
Fixed effects	Period	Period + City	Period	Period + City
Identification	Static	Static	Lagged	Lagged

+ $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

can be factored into housing prices.

Other factors that can weaken the link between residential markets and income sorting in the German housing market are institutional characteristics of the real estate market. A relatively large rental sector, lower homeownership rates, and regulatory frameworks may allow households from the entire income distribution to remain in high-demand cities, thus limiting the 'outbidding effect' and reducing the extent to which only the highest earners concentrate in these locations. In particular, an important difference between Germany and other developed countries lies in homeownership rates. With 45% Germany ranks as the country with lowest share of homeowners in the EU.[4] Voigtländer (2009) provides some arguments to explain why such a low figure. First, extensive social housing make up for a larger share of the market compared to other European countries. Second, the federal government hardly subsidizes homeowners through tax breaks or interest relief, thus reducing incentives to purchase houses. Finally, loan-to-value ratios are relatively lower compared to other large European markets, requiring buyers to provide a more substantial initial down payment.

[4] https://ec.europa.eu/eurostat/databrowser/view/ilc_lvho02__custom_12697765/bookmark/table?lang=en&bookmarkId=ae2c 4493-4b72-9f26-006774ae4eb8&c=1757069767000 retrieved on 18/02/2026

5. Conclusion

The empirical results provide only partial support for the Superstar cities effect when applied to the German context. Cities classified as Superstars exhibit systematically higher housing prices across both apartment and single-family markets. This pattern aligns with the core mechanism highlighted in the literature, whereby strong demand interacting with inelastic housing supply leads to capitalization of demand into prices rather than quantities. However, across most specifications, income differences are weak, unstable, or disappear once controlling for city fixed effects.

These findings suggest that the German housing market differs in important ways from the setting originally described by Gyourko et al. (2013). The evidence points to prices being a reflection of the value households place on access to productive labor markets, amenities, and urban opportunities, even when observed income differentials remain modest. In this sense, the results appear to be more consistent with a broader spatial equilibrium interpretation, where local characteristics are priced into real estate through willingness to pay rather than through large differences in measured income.

The analysis also highlights the importance of persistent characteristics. Once city fixed effects are included, many estimated relationships weaken substantially, indicating that structural differences across Germany play a central role in shaping housing market outcomes. Consequently, spatial divergence may be less a temporary phenomenon and more the outcome of persistent urban heterogeneity interacting with slow housing supply adjustment.

At the same time, the findings provide additional evidence that housing supply in Germany reacts only weakly to increases in demand. This is consistent with the broader literature on the elasticities of the German housing supply and helps explain why housing costs have risen so sharply in certain cities despite comparatively moderate income growth. Particularly after 2010, strong urban demand combined with limited expansion of the housing stock appears to have contributed to widening spatial disparities in prices.

Future research could extend the analysis in several directions. First, better micro-level price, income, and migration data would allow for a more direct study of household sorting across cities. Second, future work could examine the role of local regulation, zoning restrictions, and building intensity in constraining supply within German urban areas. Some attempts have already been made in Freemark (2023). Expanding knowledge on the mechanisms that drive housing supply could yield powerful insights for urban and housing policy.

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A. Appendix: Robustness Checks

A.1. Superstar Identification

For a visual account of the exclusion of COVID-19 from the sample, Figure A.1 shows the differences between the two identification methods. Compared to Figure 3.1, the overall spatial pattern of Superstar classification remains largely unchanged. Under the static identification assumption in Panel A, the same group of large cities emerges again as Superstars, and their persistent across consecutive five-year windows is not altered by the exclusion of the COVID period. Unlike the whole sample case, the progressive shift from few Superstars in the late 2000s to a larger number in the later periods is less pronounced, while the temporal concentration of Superstars toward the end of the sample remains. Importantly, no city prior to 2020 changes classification when COVID is excluded, therefore indicating that the global threshold is not strongly driven by the growth dynamics of the pandemic era. This may suggest that part of the late-sample clustering under the static definition is driven by the exceptionally strong price growth observed in the final period rather than by a steady structural expansion of Superstar markets.

Turning to the period-specific identification (Panel B), the pattern is again alike that of Figure 3.1. Superstar status is concentrated in the same markets characterized by high growth. The exclusion of COVID does not affect earlier periods as, by construction, they are a function of past growth rates.

A.2. Regression Checks

Table A.2 reports eight specifications that combine two identification strategies (Static vs. Prior Periods), two fixed effects structures (Period vs. Period plus City), and samples that include or exclude the COVID period. The main takeaway from the table is that there is no difference in significance when the years during the pandemic are excluded.

Under the static identification strategy with only period fixed effects (columns 1 and 5), the coefficient on $Superstar_i$ is positive and highly statistically significant in both samples. The estimates imply that Superstar cities exhibit apartment prices approximately 30% higher when COVID years are included and around 28% higher once they are excluded. Similarly, in the specifications including both city and period fixed effects (columns 2 and 6), the time-varying Superstar indicator remains positive and statistically significant. The estimated within-city effect is approximately 19% in the full sample and increases slightly to around 23% once the COVID period is excluded.

The lagged identification strategy yields a similar pattern for the pooled cross-sectional specifications (columns 3 and 7). The coefficient on $Superstar_i$ remains positive, economically large and highly statistically significant, with estimated price differences ranging from roughly 30% to 33% in all samples. However, once city fixed effects are introduced (columns 4 and 8), the coefficient on the time-varying Superstar indicator becomes negative and statistically insignificant in both cases. This indicates that under the lagged identification strategy, Superstar status primarily captures persistent cross-sectional differences

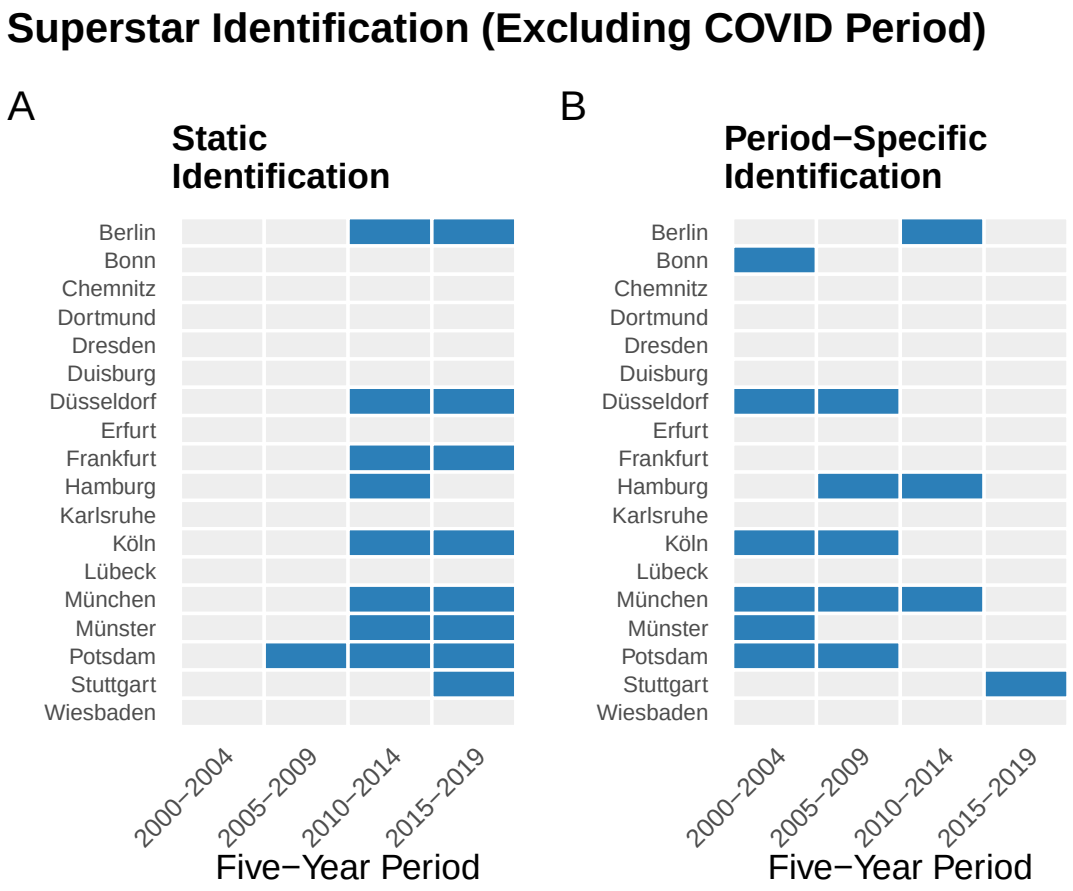

Figure A.1.: Classification of German cities as Superstar housing markets across five-year periods from 2000 to 2024. Panel A applies the static identification strategy, where inelasticity thresholds are compared with the sample median. Panel B applies period-specific identification, where thresholds are computed using two-period windows. Cities marked in blue are classified as Superstars. The training window 1995-1999 is excluded from the visualization.

across cities rather than meaningful within-city changes over time.

Overall, the robustness checks reinforce the baseline findings. The strongest and most stable within-city effects arise under the static identification strategy, consistent with Gyourko et al. (2013). Excluding the COVID period does not materially alter the results, suggesting that the observed Superstar premium reflects persistent structural features of constrained housing markets rather than temporary pandemic-related distortions.

Table A.1.: Robustness Checks for Eight Specifications: Regressing Home Value (Apartments) on Superstar

	House Value Pooled (C19)	House Value Panel (C19)	House Value Pooled (C19)	House Value Panel (C19)	House Value Pooled	House Value Panel	House Value Pooled	House Value Panel
Superstar_i	0.304***		0.327***		0.275***		0.299***	
	(0.072)		(0.072)		(0.066)		(0.065)	
Superstar_{it}		0.193**		−0.059		0.227**		−0.040
		(0.064)		(0.063)		(0.084)		(0.061)
Constant	4.312***	4.564***	4.281***	4.632***	4.328***	4.492***	4.298***	4.579***
	(0.054)	(0.034)	(0.056)	(0.061)	(0.050)	(0.027)	(0.051)	(0.056)
Num.Obs.	90	90	90	90	72	72	72	72
R2 Adj.	0.695	0.861	0.720	0.835	0.661	0.832	0.689	0.779
COVID	Included	Included	Included	Included	Excluded	Excluded	Excluded	Excluded
Identification	Static	Static	Lagged	Lagged	Static	Static	Lagged	Lagged
Fixed effects	Period	Period + City	Period	Period + City	Period	Period + City	Period	Period + City
Clustered SE	By City	By City	By City	By City	By City	By City	By City	By City

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

B. Appendix: The Pooled OLS Estimator

B.1. Interpretation of the Pooled OLS Estimator

The following derivation takes from Cameron and Trivedi (2005). Consider the pooled panel specification

$$\boldsymbol{P} = \beta \boldsymbol{S} + \boldsymbol{D}\gamma + \boldsymbol{\varepsilon},$$

where $\boldsymbol{P}$ is the stacked $(NT \times 1)$ vector of log house prices, $\boldsymbol{S}$ is the $(NT \times 1)$ vector of Superstar indicators, $\boldsymbol{D}$ is the $(NT \times T)$ matrix of period dummies, and $\boldsymbol{\varepsilon}$ is the error term. In the present application, $N = 18$ cities and $T = 6$ periods, such that $NT = 108$ observations.

To partial out common time effects, define the annihilator matrix associated with the time dummies:

$$\boldsymbol{M}_D = \boldsymbol{I}_{108} - \boldsymbol{D}(\boldsymbol{D}'\boldsymbol{D})^{-1}\boldsymbol{D}'.$$

By the Frisch-Waugh-Lovell theorem, the pooled OLS estimator for β can be written as

$$\hat{\beta} = (\boldsymbol{S}'\boldsymbol{M}_D\boldsymbol{S})^{-1}\boldsymbol{S}'\boldsymbol{M}_D\boldsymbol{P}.$$

Substituting the population model into the estimator yields

$$\hat{\beta} = (\boldsymbol{S}'\boldsymbol{M}_D\boldsymbol{S})^{-1}\boldsymbol{S}'\boldsymbol{M}_D(\beta\boldsymbol{S} + \boldsymbol{D}\gamma + \boldsymbol{\varepsilon}).$$

Using the property that $\boldsymbol{M}_D\boldsymbol{D} = 0$,

$$\begin{aligned}\hat{\beta} &= (\boldsymbol{S}'\boldsymbol{M}_D\boldsymbol{S})^{-1}\boldsymbol{S}'\boldsymbol{M}_D(\beta\boldsymbol{S} + \boldsymbol{\varepsilon}) \\ &= \beta + (\boldsymbol{S}'\boldsymbol{M}_D\boldsymbol{S})^{-1}\boldsymbol{S}'\boldsymbol{M}_D\boldsymbol{\varepsilon}.\end{aligned}$$

Taking expectations conditional on $\boldsymbol{S}$ gives

$$\mathbb{E}[\hat{\beta} \mid \boldsymbol{S}] = \beta + (\boldsymbol{S}'\boldsymbol{M}_D\boldsymbol{S})^{-1}\boldsymbol{S}'\boldsymbol{M}_D\mathbb{E}[\boldsymbol{\varepsilon} \mid \boldsymbol{S}].$$

Under the conditional mean independence assumption

$$\mathbb{E}[\boldsymbol{\varepsilon} \mid \boldsymbol{S}] = 0,$$

it follows that

$$\mathbb{E}[\hat{\beta} \mid \boldsymbol{S}] = \beta.$$

Therefore, pooled OLS consistently recovers the conditional mean difference in house prices associated with Superstar status.

However, this assumption may not hold exactly in the present setting. Since Superstar classification partly depends on past price growth and housing demand, the Superstar indicator may remain correlated with persistent unobserved local characteristics affecting

current prices, and a five-year window may be insufficient to fully control for such persistence. Consequently, the estimates are interpreted primarily as descriptive associations rather than causal effects.

B.2. Heteroskedasticity and Persistent Correlation

By construction, superstar cities are selected from the upper tail of long-run growth, implying systematic differences in residual dispersion. In particular, cities classified as Superstars may exhibit systematically larger residual variation due to stronger housing demand fluctuations, or persistent local shocks. Formally:

$$\mathrm{Var}(\varepsilon_i \mid S_i = 1) > \mathrm{Var}(\varepsilon_j \mid S_j = 0).$$

The error covariance matrix is therefore

$$\boldsymbol{\Omega} = \begin{pmatrix} \boldsymbol{\sigma}_1^2 & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\sigma}_0^2 \end{pmatrix} \neq \sigma^2 \boldsymbol{I}, \qquad \boldsymbol{\sigma}_1^2 > \boldsymbol{\sigma}_0^2.$$

Hence, OLS is not the Best Linear Unbiased Estimator due to block heteroskedasticity.

Furthermore, spatial and serial correlation may be present. This could be due to the fact that some cities within my sample, for example Potsdam and Berlin, are located close to each other. These cities are closely linked by commuters. Utility maximizer households may decide to locate in one non-superstar city to lower their housing expenditure, and then commute to work to the closest superstar city thus reaping the income benefits of higher local productivity. Moreover, growth (or decline), of cities is strongly persistent over time at it could be the case that variation in residuals also follow this persistent pattern. For these reasons, assuming that each city-observation is independent would be too strict and therefore,

$$\begin{aligned} \mathbb{E}[\varepsilon_{it}\varepsilon_{jt}] &\neq 0 \quad \text{for nearby cities,} \\ \mathbb{E}[\varepsilon_{it}\varepsilon_{i,t+1}] &\neq 0 \quad \text{within cities over time.} \end{aligned}$$

To address this issue, standard errors are clustered at the city level. Pustejovsky and Tipton (2018) provide a method, and the R package, to correct for heteroskedasticity and unmodeled dependence among the error terms. Furthermore, they propose an additional correction for small sample bias based on degrees of freedom adjustments. The correction aims to limit the likelihood of Type 1 errors in the variance of the estimates, leading to more accurate rejection rates. Indeed, my results suggest that such adjustments reduce precision of $\hat{\beta}$ once standard errors are clustered by city.

B.3. Endogeneity Concerns and Instrumental Variable Strategy

A methodological concern that could arise from this analysis is related to the validity of the pooled OLS estimator. As explained in Appendix B, by construction, the status of Superstar is determined by price growth and demand pressure in previous periods. Since current price levels P_t are a function of previous levels P_{t-5}, the assumption of mean independence (that is, $\mathbb{E}[P_{t-5}|Superstar] = \text{constant}$) is potentially violated. Specifically,

if cities are labeled Superstars based on past price growth, the Superstar dummy effectively contains the historical price trajectory of the city, making it difficult to separate the effect of the label from the city's inherent price level.

Given that geographic characteristics have already been explored to explain the variation of the supply elasticity in Beze (2023), to address the potential endogeneity of the Superstar status and the mechanical link, I employ an instrumental variable approach that uses historical data rather than topographic data. Following the methodology from Brakman et al. (2004), I use a predetermined historical shock, i. e., the share of housing destruction in German cities during WWII, as an instrument for modern housing supply inelasticity in the first stage.[1] The second stage then regresses the estimated inelasticity measure on price levels. The identification strategy rests on the premise that heavily destroyed cities had the opportunity to modernize urban layouts and expand housing supply elasticity during reconstruction, whereas historically preserved cities remained constrained by past planning. This approach serves to isolate variation in supply conditions that is exogenous to recent demand side price pressures.

However, evaluating the resulting metrics in Table B.1 reveals notable limitations with respect to the validity of this identification strategy. Although the endogeneity test statistic (0.927) suggests a persistent divergence between OLS and 2SLS estimates, the first-stage regression indicates that the WWII destruction share serves as an exceptionally weak instrument for contemporary supply inelasticity. The first-stage F-statistic = 0.149 falls significantly below the standard diagnostic threshold of 10, indicating that the instrument fails to capture sufficient exogenous variation in modern housing constraints. This lack of correlation suggests that the "rebuilding effect" has been largely diluted over time. Specifically, decades of heterogeneous post-war reconstruction trajectories, divergent urban planning regimes, and idiosyncratic local housing policies have likely decoupled current supply conditions from their historical origins. Consequently, the weak first-stage relationship precludes a definitive causal re-estimation of the Superstar effect using this historical proxy.

[1] Data for share of destruction are from Kästner (1949). Retrieved from https://leopard.tu-braunschweig.de/receive/dbbs_mods_00070003, 1949 (Vol. 37); on 14/03/2026.

Table B.1.: IV Regression Results for Housing Inelasticity and Prices

IV stages	Housing Inelasticity First (1)	log_price Second (2)
WWII Destruction	-2.295	
	(2.946)	
east	-3.831	0.5231
	(2.702)	(0.6543)
Housing Inelasticity		0.1578
		(0.1639)
Fixed-effects: **year**	Yes	Yes
F-test (1st stage), Housing Inelasticity		0.14946
Wald (2nd stage)		0.92723
Year fixed effects	✓	✓

Clustered (city) standard-errors in parentheses.
***: 0.01, **: 0.05, *: 0.1